Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · DevOps & CI/CD Automation · AI Integration · Jenkins · GitHub Actions · Terraform · Kubernetes · Java · Python

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PROFESSIONAL SUMMARY

Senior software engineer with 7+ years building production systems across the **full software development lifecycle** — backend development, **CI/CD automation, infrastructure modernization, AI integration, and DevOps engineering**. At Intuit, owns **Jenkins, GitHub Actions, and Tekton CI/CD pipelines** shipping AI-powered features to AWS via **Terraform IaC** and Kubernetes; implements **LLM orchestration and prompt engineering** for automated code generation and financial data summarization; builds **Power BI dashboards with DAX** to track platform performance and DORA metrics (Deployment Frequency, Lead Time, Change Failure Rate, MTTR). At Entrata (100K+ MAU multi-tenant SaaS), shipped 20+ features in **Java Spring Boot, Python, PHP, and React/TypeScript** across **120+ Agile sprints**; built **RESTful APIs with OAuth2 and JWT** security; architected **ETL/ELT pipelines** across 1.6M+ records; Docker microservices with **50% faster execution and 30% faster deployments**. Stack: **Jenkins, GitHub Actions, Tekton, Terraform, Kubernetes, Docker, Java Spring Boot, Python, AWS, React, TypeScript**. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

CI/CD & DevOps: Jenkins, GitHub Actions, Tekton, GitHub Enterprise, Cloud-Native CI/CD, CI/CD Pipeline Design & Optimization, Infrastructure as Code (Terraform), Kubernetes, Docker, Deployment Automation, Agile/XP, TDD, Continuous Integration, DevOps Metrics (Deployment Frequency, Lead Time, Change Failure Rate, MTTR)

Backend & APIs: Java (Spring Boot) (7+ yrs), Python (9+ yrs), Node.js, TypeScript, React, RESTful API Development, API Gateway (AWS), OAuth2 & JWT Security, GraphQL, Microservices Architecture, Async Processing (RabbitMQ), Celery

AI Integration & Data: LLM Orchestration, Prompt Engineering, Automated Code Generation, Anthropic Claude, LangGraph, LangChain, RAG Pipelines, ETL/ELT Pipelines, Power BI & DAX, Snowflake, PostgreSQL, MySQL, Redis, MongoDB, SQL Optimization

Cloud & Observability: AWS (S3, EC2, RDS, Lambda, SQS, Glue, API Gateway, IAM), GCP/Vertex AI, Prometheus, Grafana, OpenTelemetry, RBAC & Access Controls, Libsodium Encryption, HIPAA/PHI Compliance, Linux

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — CI/CD Automation · AI Integration · Developer Platform

- Own **CI/CD pipeline design and optimization** using **Jenkins, GitHub Actions, and Tekton** in the GitHub Enterprise ecosystem; manage **Terraform IaC** for consistent, auditable cloud infrastructure; deploy and manage **containerized workloads on Kubernetes** (AWS EKS) with structured logging and Prometheus/Grafana observability; track and improve **DORA metrics** (Deployment Frequency, Lead Time for Changes, Change Failure Rate, MTTR).
- Design and implement **AI integrations** using Anthropic Claude: **LLM orchestration and prompt engineering** for automated financial data summarization and natural-language query automation; build **Power BI dashboards with DAX** for platform performance and engineering KPI reporting; maintain **ETL/ELT pipelines** on Snowflake with data quality monitoring and audit trails; mentor engineers and provide technical guidance across deployments.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Senior Software Engineer — Backend Systems · Infrastructure Modernization · API Development

- Built and maintained **production RESTful APIs** in **Java Spring Boot and Python** with **OAuth2 and JWT** security for a 100K+ MAU multi-tenant platform; architected **ETL/ELT data pipelines** processing 1.6M+ HUD/Tax Credit records; integrated **enterprise APIs from HUD and Tax Credit federal data sources** with full end-to-end ownership from scoping through go-live.
- Led **Docker microservices infrastructure modernization** (50% better scalability, 30% faster deployments); built **cloud-native CI/CD pipelines** and automated deployment workflows; **AWS S3 migration** with Libsodium encryption and RBAC for PHI; **optimized 200+ SQL queries (30% performance gain)** across 1M+ records; delivered across **120+ Agile sprints** applying XP and TDD practices; reviewed 300+ PRs and mentored junior engineers.

PROJECTS

AgentProbe — Open-Source Agent Runtime with CI/CD Regression Harness

CI/CD · AI Integration · Automation

Python · FastAPI · LangGraph · GitHub Actions CI/CD · ChromaDB · Anthropic · OpenAI · Groq — github.com/venkatalakshmiparimala/AgentProbe

- **LLM orchestration platform** with 15-endpoint FastAPI backend; **GitHub Actions CI/CD regression harness** with golden datasets running on every commit; **prompt engineering** for automated code generation and validation; 3 LangGraph agents with tool calling; Docker containerized; ships to real users with **zero external dependencies**.

Leasel — Production AI System with Full CI/CD Pipeline

LLM Orchestration · CI/CD · Cloud Deployment

Python · LangGraph · FastAPI · GitHub Actions · Docker · AWS — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production AI system deployed via **automated CI/CD pipeline** (GitHub Actions → Docker → AWS); **LLM orchestration** across 8-agent LangGraph workflow with prompt engineering, guardrails, and structured outputs; CI eval harness (LLM-as-judge + golden datasets): **93.6% precision, 87.9% F1** across 1,240 test cases; full observability with structured logging.

AI on FHIR — Full-Stack API with OAuth2/JWT Security

API Security · OAuth2 · Full-Stack

Python · Flask · React · TypeScript · Docker · REST APIs · OAuth2 · JWT — github.com/venkatalakshmiparimala/AI_on_FHIR

- Full-stack application (React/TypeScript + Python/Flask) with **RESTful API development, OAuth2 and JWT authentication, and HIPAA-style RBAC**; Docker containerized with automated deployment; **API Gateway integration** with fine-grained access controls; cut manual clinical lookup time ~60%.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Distributed systems, ML pipelines, cloud infrastructure, data governance, software architecture

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017